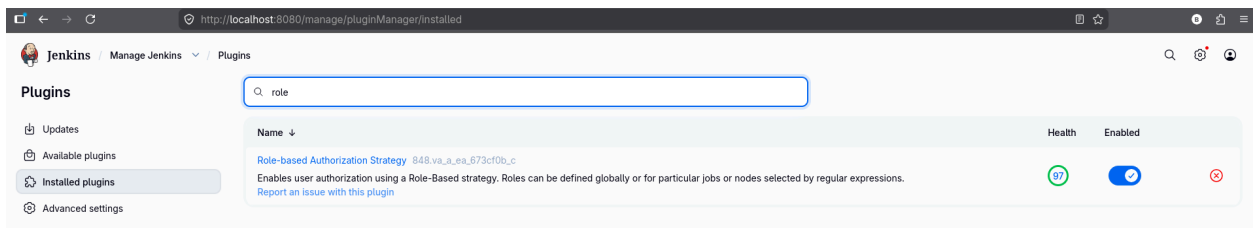


Name: Bassant Ali Kamal Ali
Track: Cloud Platform Development
Course: Jenkins
Lab 1

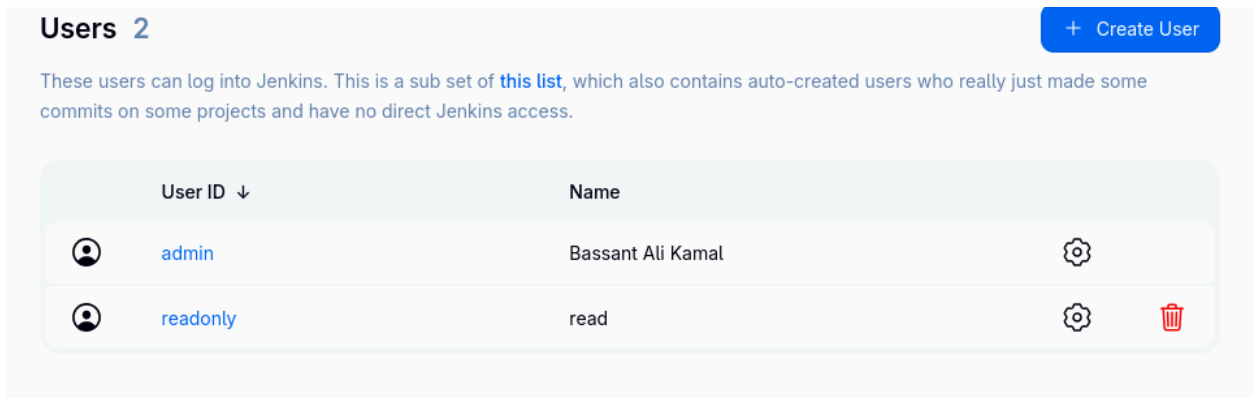
1- install jenkins with docker image

```
bassant@fedora:~$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS
b80de3c666d6   jenkins/jenkins:lts   "/usr/bin/tini -- /u...  7 hours ago   Up 6 hours   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:5000
0->50000/tcp, [::]:50000->50000/tcp
bassant@fedora:~$
```

2- install role based authorization plugin



3- create new user



4- create read role and assign it to the new user

First option:

Authorization

Matrix-based security

User/group	Overall		Credentials		Agent					Job					Run		View		SCM		Metrics																
	Administer	Manage	Read	Create	Delete	ManageDomains	Update	View	Build	Configure	Connect	Create	Delete	Disconnect	Provision	Build	Cancel	Configure	Create	Delete	Discover	Move		Read	Workspace	Delete	Replay	Update	Configure	Create	Delete	Read	Tag	HealthCheck	ThreadDump	View	
Anonymous	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☒	☐	
Authenticated Users	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☒	☐	
Bassant Ali Kamal	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☒	☐	
read	☐	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☒	☐	
Add user...		Add group...																																			

Second option:

Jenkins

Manage Jenkins

Manage and Assign Roles

Manage Roles

Manage Roles

Assign Roles

Permission Templates

Role Strategy Macros

Manage Roles

Global roles

Role	Overall		Credentials			Agent					Job					Run		View			SCM		Metrics												
	Administer	Manage	Read	Create	Delete	ManageDomains	Update	View	Build	Configure	Connect	Create	Delete	Disconnect	Provision	Build	Cancel	Configure	Create	Delete	Discover	Move	Read	Workspace	Delete	Replay	Update	Configure	Create	Delete	Read	Tag	HealthCheck	ThreadDump	View
 admin	☒																																		
 read		☒																																	

← → ↻ http://localhost:8080/manage/role-strategy/assign-ro

 **Jenkins** / Manage Jenkins ▾ / Manage and Assign Roles / Assign Roles

 Manage Roles

 **Assign Roles**

 Permission Templates

 Role Strategy Macros

Assign Roles

Global roles

User/Group	admin	read	
 Anonymous	☐	☐	
 Authenticated Users	☐	☐	
  Bassant Ali Kamal	☒	☐	
  read	☐	☒	

5- create free style pipeline and link it to private git repo(inside it create directory and create file with "hello world")

 General

 **Source Code Management**

 Triggers

 Environment

 Build Steps

 Post-build Actions

☐ None

☒ **Git** ?

Repositories ?

Repository URL ?

Credentials ?

Advanced ▾

⚙️ General

🔗 Source Code Management

🕒 Triggers

🌐 Environment

📋 Build Steps

📦 Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

⋮ Execute shell ?

Command

See [the list of available environment variables](#)

```
mkdir demo
cd demo
echo "hello world" > hello.txt
ls
cat hello.txt
```

Advanced ▾

🏠 ⬅ ➡ ↻

🔒 http://localhost:8080/job/lab1-freestyle/

Show sidebar (Alt+Ctrl+Z)

 **Jenkins** / lab1-freestyle

📄 Status

</> Changes

📁 Workspace

▶ Build Now

⚙️ Configure

🗑 Delete Project

✎ Rename

✅ lab1-freestyle

Permalinks

- [Last build \(#2\), 48 sec ago](#)
- [Last stable build \(#2\), 48 sec ago](#)
- [Last successful build \(#2\), 48 sec ago](#)
- [Last failed build \(#1\), 2 min 4 sec ago](#)
- [Last unsuccessful build \(#1\), 2 min 4 sec ago](#)
- [Last completed build \(#2\), 48 sec ago](#)

Builds > ⋮

🔍 Filter

/

Today

✅ #2 6:43 PM ▾

❌ #1 6:42 PM ▾

http://localhost:8080/job/lab1-freestyle/2/

Jenkins / lab1-freestyle / #2

Status

</> Changes

Console Output

Edit Build Information

Delete build '#2'

Timings

Git Build Data

Previous Build

✓ #2 (May 9, 2026, 6:43:54 PM)

Started by user Bassant Ali Kamal

This run spent:

- 6 ms waiting;
- 4.3 sec build duration;
- 4.3 sec total from scheduled to completion.

Revision: c59f1a8ea629043f896381b626745055d302d436

Repository: https://github.com/Besso2003/Jenkins_ITI.git

- refs/remotes/origin/main

</> No changes.

http://localhost:8080/job/lab1-freestyle/2/console

Jenkins / lab1-freestyle / #2

Status

</> Changes

Console Output

Edit Build Information

Delete build '#2'

Timings

Git Build Data

Previous Build

Console

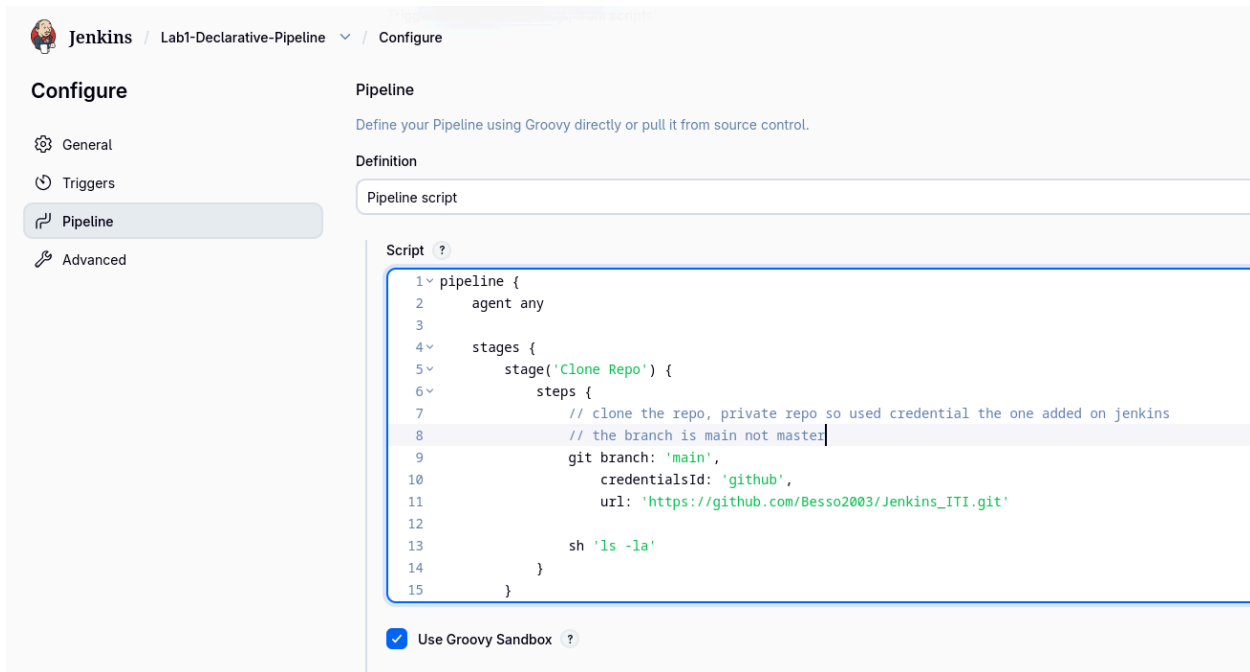
```
Started by user Bassant Ali Kamal
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/lab1-freestyle
[WS-CLEANUP] Deleting project workspace...
[WS-CLEANUP] Deferred wipeout is used...
[WS-CLEANUP] Done
The recommended git tool is: NONE
using credential github
Cloning the remote Git repository
Cloning repository https://github.com/Besso2003/Jenkins_ITI.git
> git init /var/jenkins_home/workspace/lab1-freestyle # timeout=10
Fetching upstream changes from https://github.com/Besso2003/Jenkins_ITI.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/Besso2003/Jenkins_ITI.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/Besso2003/Jenkins_ITI.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision c59f1a8ea629043f896381b626745055d302d436 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f c59f1a8ea629043f896381b626745055d302d436 # timeout=10
Commit message: "Revise Lab 1 objectives for Jenkins basics"
First time build. Skipping changelog.
[lab1-freestyle] $ /bin/sh -xe /tmp/jenkins16226396411146133136.sh
+ mkdir demo
+ cd demo
+ echo hello world
+ ls
hello.txt
+ cat hello.txt
hello world
Finished: SUCCESS
```

```

bassant@fedora:~$ docker exec -it jenkins bash
jenkins@b80de3c666d6:/$ cd /var/jenkins_home/workspace/
jenkins@b80de3c666d6:~/workspace$ ls
lab1-freestyle  lab1-freestyle@tmp
jenkins@b80de3c666d6:~/workspace$ cd lab1-freestyle
jenkins@b80de3c666d6:~/workspace/lab1-freestyle$ ls
README.md  demo
jenkins@b80de3c666d6:~/workspace/lab1-freestyle$ cd demo
jenkins@b80de3c666d6:~/workspace/lab1-freestyle/demo$ ls
hello.txt
jenkins@b80de3c666d6:~/workspace/lab1-freestyle/demo$ cat hello.txt
hello world
jenkins@b80de3c666d6:~/workspace/lab1-freestyle/demo$

```

6- create declarative in jenkins GUI pipeline for your own repo to do "ls"



The screenshot shows the Jenkins web interface for configuring a pipeline. The left sidebar has a 'Configure' section with sub-links: General, Triggers, Pipeline (selected), and Advanced. The main content area is titled 'Pipeline' and includes a description: 'Define your Pipeline using Groovy directly or pull it from source control.' Below this is a 'Definition' section with a 'Pipeline script' tab. The script is displayed in a text editor with line numbers 1 through 15. At the bottom, there is a checkbox labeled 'Use Groovy Sandbox' which is checked.

```

1 pipeline {
2   agent any
3
4   stages {
5     stage('Clone Repo') {
6       steps {
7         // clone the repo, private repo so used credential the one added on jenkins
8         // the branch is main not master:
9         git branch: 'main',
10            credentialsId: 'github',
11            url: 'https://github.com/Besso2003/Jenkins_ITI.git'
12
13         sh 'ls -la'
14       }
15     }
16   }
17 }

```

☒ Use Groovy Sandbox



Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces



#1 (May 9, 2026, 7:48:23 PM)



Started by user [Bassant Ali Kamal](#)



This run spent:

- 7 ms waiting;
- 3.3 sec build duration;
- 3.3 sec total from scheduled to completion.



git

Revision: c59f1a8ea629043f896381b626745055d302d436

Repository: https://github.com/Besso2003/Jenkins_ITI.git

- refs/remotes/origin/main



No changes.

Jenkins / Lab1-Declarative-Pipeline / #1

Changes

Console Output

- Edit Build Information
- Delete build '#1'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

```
Started by user Bassant Ali Kamal
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/jenkins_home/workspace/Lab1-Declarative-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repo)
[Pipeline] git
The recommended git tool is: NONE
using credential github
Cloning the remote Git repository
Cloning repository https://github.com/Besso2003/Jenkins_ITI.git
> git init /var/jenkins_home/workspace/Lab1-Declarative-Pipeline # timeout=10
Fetching upstream changes from https://github.com/Besso2003/Jenkins_ITI.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/Besso2003/Jenkins_ITI.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/Besso2003/Jenkins_ITI.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision c59f1a8ea629043f896381b626745055d302d436 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f c59f1a8ea629043f896381b626745055d302d436 # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git checkout -b main c59f1a8ea629043f896381b626745055d302d436 # timeout=10
Commit message: "Revise Lab 1 objectives for Jenkins basics"
First time build. Skipping changelog.
[Pipeline] sh
+ ls -la
total 4
drwxr-xr-x. 1 jenkins jenkins 26 May  9 19:48 .
drwxr-xr-x. 1 jenkins jenkins 260 May  9 19:48 ..
drwxr-xr-x. 1 jenkins jenkins 136 May  9 19:48 .git
-rw-r--r--. 1 jenkins jenkins 551 May  9 19:48 README.md
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

7- create scripted in jenkins GUI pipeline for your own repo to do "ls"

Jenkins / Lab1-Script-Pipeline / Configure

Configure

- General
- Triggers
- Pipeline
- Advanced

General

Description

Plain text [Preview](#)

☐ Discard old builds ?

☒ Do not allow concurrent builds

☐ Abort previous builds ?

 **Jenkins** / Lab1-Script-Pipeline / Configure

Configure

 General

 Triggers

 **Pipeline**

 Advanced

Definition

Pipeline script

Script ?

```
1 node {
2   def mvnHome
3   stage('Clone Repo') { // for display purposes
4     // Get some code from a GitHub repository
5     // branch name is main not master so i had to write it
6     // the repo is private so i had to also write the Credentials that i added in jenkins "github"
7     git credentialsId: 'github',
8       branch: 'main',
9       url: 'https://github.com/Besso2003/Jenkins_ITI.git'
10  }
11  stage('List Files') {
12    // Run ls -la
13    sh 'ls -la'
14  }
15 }
```

☒ Use Groovy Sandbox ?

 **Jenkins** / Lab1-Script-Pipeline

 **Status**

 Changes

 Build Now

 Configure

 Delete Pipeline

 Stages

 Rename

 Pipeline Syntax

 **Lab1-Script-Pipeline**

Permalinks

- [Last build \(#2\), 1 min 36 sec ago](#)
- [Last stable build \(#2\), 1 min 36 sec ago](#)
- [Last successful build \(#2\), 1 min 36 sec ago](#)
- [Last failed build \(#1\), 3 min 41 sec ago](#)
- [Last unsuccessful build \(#1\), 3 min 41 sec ago](#)
- [Last completed build \(#2\), 1 min 36 sec ago](#)

Builds > ...



Today

 **#2** 7:28 PM

 **#1** 7:26 PM



Status

- Changes
- Console Output
- Edit Build Information
- Delete build '#2'
- Timings
- Git Build Data
- Pipeline Overview
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

#2 (May 9, 2026, 7:28:24 PM)



Started by user [Bassant Ali Kamal](#)



This run spent:

- 7 ms waiting;
- 1.7 sec build duration;
- 1.7 sec total from scheduled to completion.



Revision: c59f1a8ea629043f896381b626745055d302d436

Repository: https://github.com/Besso2003/Jenkins_ITI.git

- refs/remotes/origin/main



No changes.

Jenkins / Lab1-Script-Pipeline #2

Status

Changes

Console Output

Edit Build Information

Delete build '#2'

Timings

Git Build Data

Pipeline Overview

Replay

Pipeline Steps

Workspaces

Previous Build

Console

```
Started by user Bassant Ali Kamal
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/jenkins_home/workspace/Lab1-Script-Pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repo)
[Pipeline] git
The recommended git tool is: NONE
using credential github
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/Lab1-Script-Pipeline/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/Besso2003/Jenkins_ITI.git # timeout=10
Fetching upstream changes from https://github.com/Besso2003/Jenkins_ITI.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/Besso2003/Jenkins_ITI.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision c59f1a8ea629043f896381b626745055d302d436 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f c59f1a8ea629043f896381b626745055d302d436 # timeout=10
> git branch -a -v --no-abbrev # timeout=10
> git checkout -b main c59f1a8ea629043f896381b626745055d302d436 # timeout=10
Commit message: "Revise Lab 1 objectives for Jenkins basics"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (List Files)
[Pipeline] sh
+ ls -la
total 4
drwxr-xr-x. 1 jenkins jenkins 26 May  9 19:28 .
drwxr-xr-x. 1 jenkins jenkins 152 May  9 19:26 ..
drwxr-xr-x. 1 jenkins jenkins 136 May  9 19:28 .git
-rw-r--r--. 1 jenkins jenkins 551 May  9 19:28 README.md
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Here i did jenkinsfile in branch main but then i would add the file in each branch which is not best approach

Files

main

+

Q

Q Go to file T

Jenkinsfile

README.md

Jenkins_ITI / Jenkinsfile

Besso2003

 Add Jenkins pipeline to list files

Code

Blame

 11 lines (10 loc) · 149 Bytes

```
1 pipeline {
2     agent any
3
4     stages {
5         stage('List Files') {
6             steps {
7                 sh 'ls -la'
8             }
9         }
10    }
11 }
```

Jenkins / Lab1-Pipeline-SCM / Configure

Configure

General

Triggers

Pipeline

Advanced

Definition

Pipeline script from SCM

SCM ?

Git

Repositories ?

Repository URL ?

https://github.com/Besso2003/Jenkins_ITI.git

Credentials ?

Besso2003/*****

+ Add

Advanced

+ Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

main



Jenkins

Lab1-Pipeline-SCM

#1

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Pipeline Overview

Restart from Stage

Replay

Pipeline Steps

Workspaces



#1 (May 9, 2026, 7:58:07 PM)



Started by user [Bassant Ali Kamal](#)



This run spent:

- 12 ms waiting;
- 3.9 sec build duration;
- 3.9 sec total from scheduled to completion.



git

Revision: be73dd235725b67fbe78438500db3550a49e8e6c

Repository: https://github.com/Besso2003/Jenkins_ITI.git

- origin/main



No changes.

The screenshot shows the Jenkins console output for a pipeline named 'Lab1-Pipeline-SCM'. The output details the steps taken to clone a repository, checkout a specific commit, and list files. The pipeline is running on a Jenkins node in the workspace '/var/jenkins_home/workspace/Lab1-Pipeline-SCM'. The output shows the following steps:

```
Started by user Bassant Ali Kamal
Obtained Jenkinsfile from git https://github.com/Besso2003/Jenkins_ITI.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/jenkins_home/workspace/Lab1-Pipeline-SCM
[Pipeline] {
[Pipeline] stage
[Pipeline] (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential github
Cloning the remote Git repository
Cloning repository https://github.com/Besso2003/Jenkins_ITI.git
> git init /var/jenkins_home/workspace/Lab1-Pipeline-SCM # timeout=10
Fetching upstream changes from https://github.com/Besso2003/Jenkins_ITI.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/Besso2003/Jenkins_ITI.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/Besso2003/Jenkins_ITI.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse origin/main^(commit) # timeout=10
Checking out Revision be73dd235725b67f7be7843850db3550a49e8e6c (origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f be73dd235725b67f7be7843850db3550a49e8e6c # timeout=10
Commit message: "Add Jenkins pipeline to list files"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] (List Files)
[Pipeline] sh
+ ls -la
total 8
drwxr-xr-x. 1 jenkins jenkins 48 May  9 19:58 .
drwxr-xr-x. 1 jenkins jenkins 336 May  9 19:58 ..
drwxr-xr-x. 1 jenkins jenkins 136 May  9 19:58 .git
-rw-r--r--. 1 jenkins jenkins 149 May  9 19:58 Jenkinsfile
-rw-r--r--. 1 jenkins jenkins 551 May  9 19:58 README.md
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

8- create the same with jenkinsfile in your branches as multibranch pipeline

The screenshot shows the Jenkins Build Queue and Build History. The Build Queue is empty, and the Build History shows a list of builds. The table below represents the data shown in the Build History table.

S	W	Name ↓	Last Success	Last Failure	Last Duration
✓	☀	Lab1-Declarative-Pipeline	29 min #1	N/A	3.3 sec
✓	☁	lab1-freestyle	1 hr 33 min #2	1 hr 35 min #1	4.3 sec
🔗	☀	Lab1-Multibranch-Pipeline	31 sec log	N/A	3 sec
✓	☀	Lab1-Pipeline-SCM	19 min #1	N/A	3.9 sec
✓	☁	Lab1-Script-Pipeline	49 min #2	51 min #1	1.7 sec

Lab1-Multibranch-Pipeline

Status

Configure

Scan Multibranch Pipeline Now

Scan Multibranch Pipeline Log

Multibranch Pipeline Events

Delete Multibranch Pipeline

Build History

Lab1-Multibranch-Pipeline

Branches (2)

S	W	Name ↓	Last Success	Last Failure	Last Duration	
		dev	N/A	N/A	N/A	
		master	N/A	N/A	N/A	

Lab1-Multibranch-Pipeline

Scan Multibranch Pipeline

Configure

Scan Multibranch Pipeline Now

Scan Multibranch Pipeline Log

View as plain text

Multibranch Pipeline Events

Delete Multibranch Pipeline

Build History

Project Relationship

Check File Fingerprint

Rename

Pipeline Syntax

Credentials

Build Queue

No builds in the queue.

Build Executor Status

0/2

```

Started by user Bassant Ali Kamal
[Sat May 09 20:17:09 UTC 2026] Starting branch indexing...
> git --version # timeout=10
> git --version # 'git version 2.47.3'
using GIT_ASKPASS to set credentials
> git ls-remote --symref -- https://github.com/Besso2003/Jenkins_ITI.git # timeout=10
> git rev-parse --resolve-git-dir /var/jenkins_home/caches/git-c72dfba5bda93c282a6e46272c09e24b/.git # timeout=10
Setting origin to https://github.com/Besso2003/Jenkins_ITI.git
> git config remote.origin.url https://github.com/Besso2003/Jenkins_ITI.git # timeout=10
Fetching & pruning origin...
Listing remote references...
> git config --get remote.origin.url # timeout=10
> git --version # timeout=10
> git --version # 'git version 2.47.3'
using GIT_ASKPASS to set credentials
> git ls-remote -h -- https://github.com/Besso2003/Jenkins_ITI.git # timeout=10
Fetching upstream changes from origin
> git config --get remote.origin.url # timeout=10
using GIT_ASKPASS to set credentials
> git fetch --no-tags --force --progress --prune -- origin +refs/heads/*:refs/remotes/origin/* # timeout=10
Checking branches...
Checking branch master
'Jenkinsfile' found
Met criteria
Did not schedule build for branch: master
Checking branch dev
'Jenkinsfile' found
Met criteria
Did not schedule build for branch: dev
Processed 2 branches
[Sat May 09 20:17:12 UTC 2026] Finished branch indexing. Indexing took 3 sec
Finished: SUCCESS

```